2026 NASA INTERNATIONAL SPACE APPS CHALLENGE

INTERIM

ASTRONAUT HEALTH ASSESSMENT & DECISION-MAKING SIMULATOR

Selected challenge:

Create Health Monitoring Software for Astronauts on Space Missions

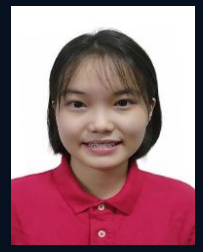
Making health decisions when communication with Earth is delayed.

LIVE DEMO https://limlining.github.io/Nasa-Interim/ • **GITHUB** https://github.com/limlining/Nasa-Interim

THE IDEA

A web-based educational simulator where users experience astronaut health scenarios, assess simulated information and practise choosing a response.

TEAM A++


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Why astronaut health decisions are difficult

Spaceflight hazards can affect health, while distance from Earth can delay medical guidance.

SPACE RADIATION

ISOLATION & CONFINEMENT

DISTANCE FROM EARTH

GRAVITY FIELDS

HOSTILE / CLOSED ENVIRONMENTS

THE CHALLENGE

How can astronauts gather health information, assess their condition and take timely action when immediate ground support is limited?

OUR RESPONSE

INTERIM provides a scenario-based simulator to practise health assessment and decisions under delayed or unavailable communication.

NASA reference: Human Research Program — Five Hazards of Human Spaceflight

How NASA research informs INTERIM

NASA open research helps define what to monitor and why it matters during spaceflight.

NASA STANDARD MEASURES

Research areas include activity, sleep, cardiovascular function, immune response and sensorimotor performance.

SPACE-BASED TECHNOLOGY

Wearable sensors and spacecraft health-monitoring technologies can collect physiological and behavioural measurements during missions.

HOW WE APPLY THE RESEARCH

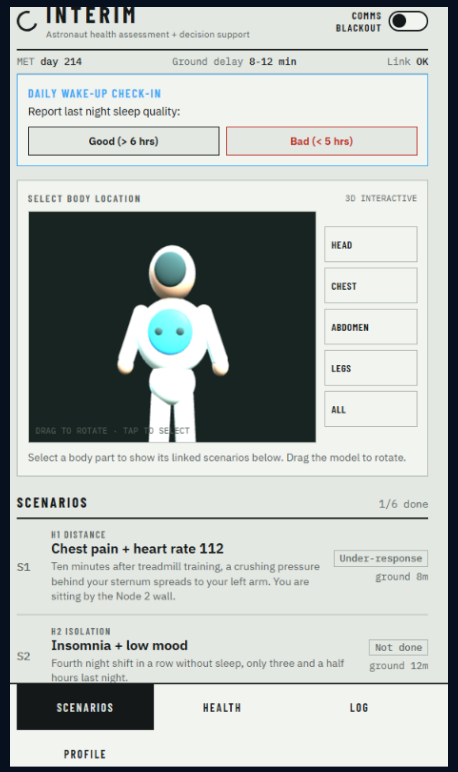
These research themes informed our scenario choices, simulated indicators, daily check-in and health-assessment workflow.

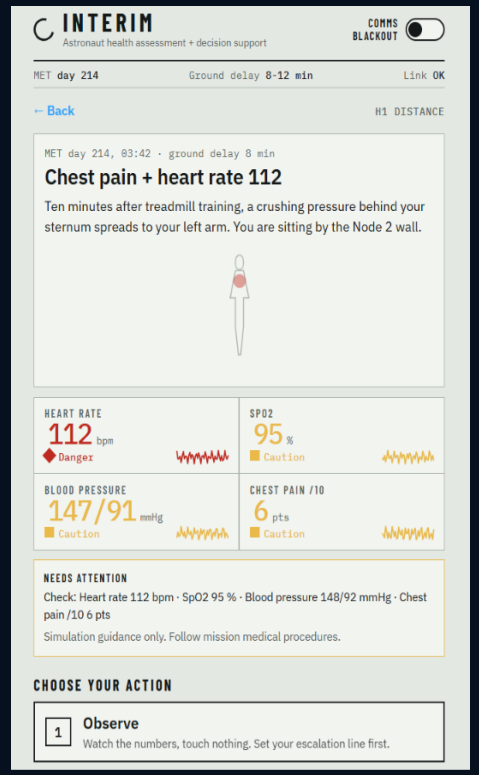
Current prototype: simulated values only. No NASA raw dataset, astronaut medical record, live spacecraft feed or wearable sensor is integrated.

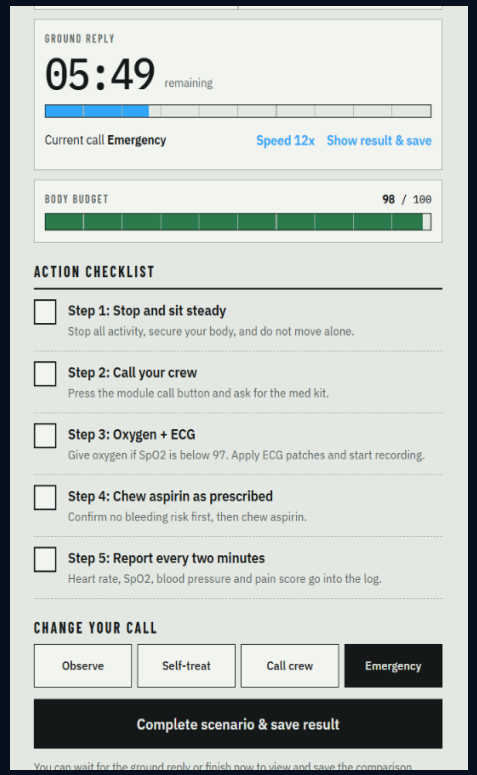
NASA references: nasa.gov/hrp/hazards/ • nasa.gov/hrp/risks/ • nasa.gov/hrp/ongoing-hrp-research/

What users experience

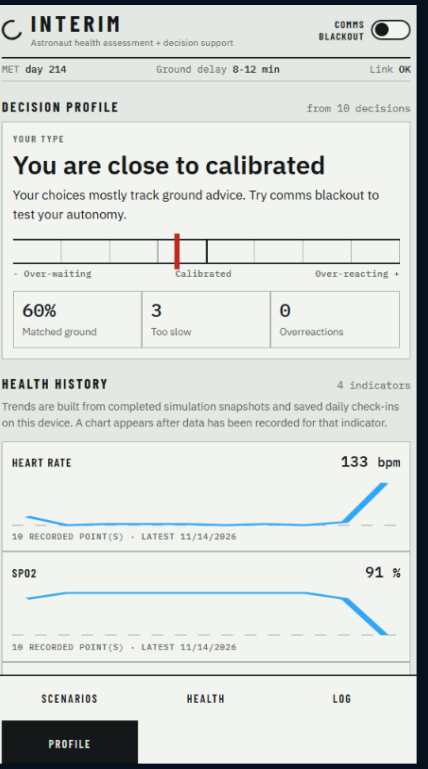
Review the situation, assess simulated health information and choose a response.











01 ASSESS

Symptoms, indicators and trends

02 DECIDE

Observe • Self-treat • Call crew • Emergency

03 REFLECT

Feedback, Decision Log and progress profile

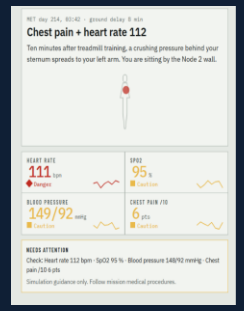
Illustrative interface preview

Six simulated astronaut health scenarios

A range of physical and psychological situations for guided practice.

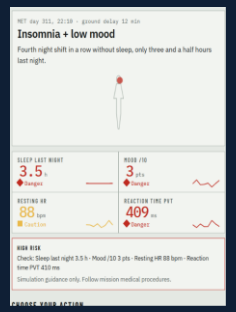
01 CHEST PAIN

Chest pain + elevated heart rate



02 INSOMNIA & LOW MOOD

Sleep and mood changes



03 ELEVATED CO_2

High CO_2 + headache



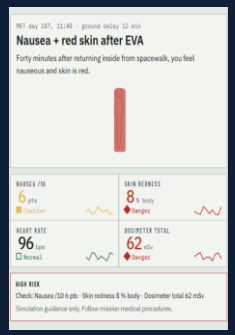
04 LOW BLOOD PRESSURE

Dizziness + low blood pressure



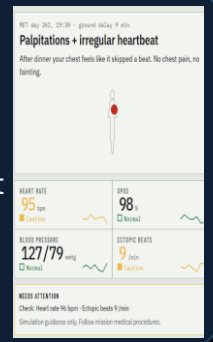
05 POST-EVA SYMPTOMS

Nausea + skin redness



06 IRREGULAR HEARTBEAT

Palpitations + irregular heartbeat

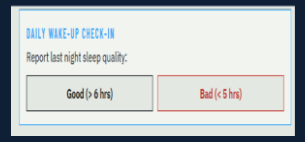


Communication Delay and Blackout modes simulate limited ground contact. Daily check-in, body navigation and Decision Log support the learning journey.

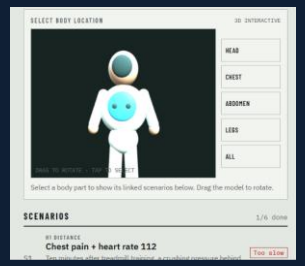
From check-in to reflection

The user moves through a repeatable decision loop.

1 CHECK IN



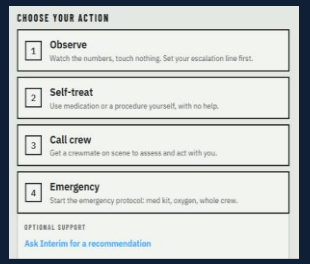
2 SELECT



3 ASSESS



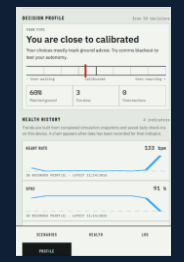
4 DECIDE



5 WAIT / ACT



6 REVIEW



TIME-AWARE PRACTICE

The simulator evaluates decisions against scenario timing and simulated guidance.

LEARNING RECORD

Users revisit decisions and review progress after completing scenarios.

Why it matters — and what comes next

An educational prototype, not a clinical monitoring or treatment system.

INTENDED BENEFIT

- Space-health awareness
- Assessment practice
- Timely action awareness
- Reflection and learning

CURRENT SCOPE

- Simulated health values
- No live sensors or NASA data feed
- No clinically validated rules
- Not for diagnosis or treatment

NEXT STEPS

- Explore public NASA datasets
- Assess approved wearable integration
- Add scenarios and indicators
- Seek expert validation

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NASA research informed the design; all values shown in the prototype are simulated.